

Ingredients required for a complete analysis of the Statistical System

1) Specification of the state of the system:

- Outcomes need to be properly expressed

2) Statistical ensemble:

- Describe the system in terms of probabilities
- Instead of paying attention to a single exp. we consider the ensemble of many.

3) Basic postulate about a priori probabilities:

- Ex: Postulate that all such combinations are equally possible.
- Experiment will show whether or not these postulates hold.

4) Probability Calculations:

* Dice analogy will lead $\rightarrow$ Many particle systems

Specification of the state of the system:

* System described by a wave ψ^N .

$\psi(q_1, \dots, q_f)$ | $\left\{ \begin{array}{l} f: \# \text{ of D.O.F of the} \\ \text{system} \end{array} \right\}$

- A quantum state is then given by giving the values of some set of f -quantum numbers.

* If $\psi(q_1, \dots, q_f)$ is given at a time t , then we can determine the dynamics via

$$|\psi(t, q_1, \dots, q_f)\rangle = U(t, 0) |\psi_0\rangle$$

$$U(t, 0) = e^{-i\hat{H}t}$$

ex 2 |: Single Particle: fixed at some q .

$$\omega \quad S = \frac{1}{2} \quad \text{Intrinsic A.M.} = \frac{1}{2} \hbar$$

$|\psi\rangle$ is characterised by the projection of m of its spin along the fixed z -axis

$$m = \frac{1}{2} \quad \text{or} \quad m = -\frac{1}{2}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} |+\rangle + \frac{1}{\sqrt{2}} |-\rangle$$

$$L_z |\psi\rangle = \frac{1}{2} \hbar |\psi\rangle$$

ex 2 |: N-particle state: All at fixed $q_{(n)}$

each with spin $\frac{1}{2}$

$$m = \frac{1}{2} \quad \text{or} \quad m = -\frac{1}{2}$$

i.e. each may have different helicity

ex 3 | 1D Harmonic Oscillator:

$$i\partial_t \psi = \hat{H}\psi$$

$$H = \frac{\hat{p}^2}{2\mu} + \frac{1}{2}\omega \hat{x}^2 = -\frac{\partial^2}{\partial x^2} + \frac{1}{2}\omega \hat{x}^2$$

at $t=0$:

$$\hat{H}\psi = E\psi$$

$$-\frac{1}{2\mu} \frac{\partial^2}{\partial x^2} \mathcal{X}(x) + \frac{1}{2}\omega \hat{x}^2 \mathcal{X}(x) = E \mathcal{X}(x)$$

$$-\frac{1}{2\mu} \partial_x^2 \mathcal{X}(x) + \frac{1}{2}\omega x^2 \mathcal{X} - E \mathcal{X}(x) = 0$$

$$\partial^2 \mathcal{X}$$

* Note $\frac{\partial^2}{\partial x^2} - x^2 \chi = 0$ has a solution of the form $\sim e^{-x^2/2}$
 - sign since $|\psi|^2 \rightarrow 0$ as $x \rightarrow \infty$
 i.e. Compact Support.

$$\chi(x) = h(x) e^{-x^2/2}$$

$$\begin{aligned} \Rightarrow -\frac{\partial}{\partial x} \left(h_x e^{-x^2/2} - h(x) x e^{-x^2/2} \right) \\ = -h_{xx} e^{-x^2/2} + h_x x e^{-x^2/2} + h_x(x) x e^{-x^2/2} \\ + h(x) e^{-x^2/2} - h(x) x^2 e^{-x^2/2} \\ = -h_{xx} e^{-x^2/2} + 2x \frac{\partial h}{\partial x} e^{-x^2/2} + h(x) e^{-x^2/2} \\ - h(x) x^2 e^{-x^2/2} \end{aligned}$$

$$\begin{aligned} \Rightarrow \frac{\partial^2 h}{\partial x^2} - 2x \frac{\partial h}{\partial x} - h(x) + h(x) x^2 - \frac{1}{2} x^2 h(x) \\ + E h(x) = 0 \end{aligned}$$

$$\therefore \frac{\partial^2 h}{\partial x^2} - 2x \frac{\partial h}{\partial x} + h(x) x^2 + h(x) = 1$$

$$\frac{\partial^2}{\partial x^2} \chi = x^2 \chi(x) + \epsilon \chi(x) = 0$$

$$\frac{\partial}{\partial x} \left[h_x e^{-x^2/2} - x h(x) e^{-x^2/2} \right] - x^2 h(x) e^{-x^2/2} + \epsilon h(x) e^{-x^2/2}$$

$$= h_{xx} e^{-x^2/2} - h_x x e^{-x^2/2} - h(x) e^{-x^2/2} - x h_x e^{-x^2/2} \\ + h(x) x^2 e^{-x^2/2} - h(x) x^2 e^{-x^2/2} + \epsilon h(x) e^{-x^2/2}$$

$$= \frac{\partial^2 h}{\partial x^2} - 2x \frac{\partial h}{\partial x} + (\epsilon - 1) h(x) = 0$$

* Hermite polynomials: $H_n(x)$ defined by the generating functions:

$$e^{t(x+t)} = e^{-t^2 + 2tx} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$$

$$(i) \quad H_0(x) + H_1(x) \frac{t}{1!} + H_2(x) \frac{t^2}{2!} + \dots$$

$$\Rightarrow H_0(x) = 1$$

$$g(0, t) = e^{-t^2} = \sum_{n=0}^{\infty} H_n(0) \frac{t^n}{n!}$$

but

$$e^{-t^2} = \sum_{n=0}^{\infty} \frac{(-t^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{n!}$$

• let $2n = m$, then

$$e^{-t^2} = \sum_{n=0}^{\infty} \frac{(-1)^{n/2} t^n}{(\frac{n}{2})!} = \sum_{n=0}^{\infty} \left((-1)^{n/2} \frac{n!}{(n/2)!} \right) \frac{t^n}{n!}$$

$$\therefore H_n(0) = (-1)^{n/2} \frac{n!}{(n/2)!}$$

$$H_n(-x) = (-1)^n H_n(x)$$

$$H_{n+1}(x) = 2x H_n(x) - 2n H_{n-1}(x)$$

$$\hookrightarrow H_n''(x) - 2x H_n'(x) + 2n H_n(x) = 0$$

$$E - 1 = 2n$$

$$\therefore \psi(x) = C_n H_n \left(x \sqrt{\frac{m\omega}{\hbar}} \right) e^{-m\omega x^2 / 2\hbar}$$

$$\star \langle x_n, x_m \rangle = \int C_n C_m H_n H_m e^{-x^2} = \pi^{1/2} 2^n n! \delta_{nm}$$

$$|C_n| = \sqrt{\frac{1}{\pi^{1/2} 2^n n!}}$$

$$= |C_n|^2 \pi^{1/2} 2^n n! = 1$$

$$\psi_n = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} H_n \left(\frac{x}{l_0} \right) e^{-x^2 / 2l_0^2}$$

$$E_n = (n + \frac{1}{2}) \hbar \omega$$

* 3D Box: (No spin) $0 \leq x_i \leq L_i$

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2$$

$$\Rightarrow -\frac{\hbar^2}{2\mu} \nabla^2 \psi = E \psi$$

* Sep of variables:

$$-\frac{\hbar^2}{2\mu} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} \right) = E$$

$$E_x + E_y + E_z = E$$

$$\frac{\partial^2 X}{\partial x^2} = -X \lambda_x$$

$$X(x) = A \cos(\lambda_x x) + B \sin(\lambda_x x)$$

B.C.: 4 variables along the Box

$$X(0) = 0 \Rightarrow A = 0$$

$$X(x) = B \sin(\lambda_x x)$$

$$X(L_x) = B \sin(\lambda_x L_x) = 0$$

$$\lambda_x L_x = n\pi \Rightarrow$$

$$\lambda_x = \frac{n\pi}{L_x}$$

$$X(x) = B \sin \left(x \frac{n_x \pi}{L_x} \right)$$

Similarly for all others

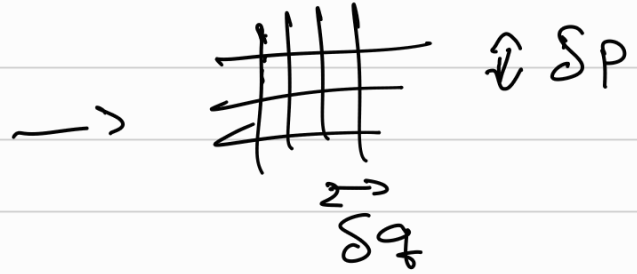
$$\psi(x, y, z) = C \sin \dots$$

* Energy: $\frac{\hbar^2}{2\mu} \left(\left(\frac{n_x \pi}{L_x} \right)^2 + \left(\frac{n_y \pi}{L_y} \right)^2 + \left(\frac{n_z \pi}{L_z} \right)^2 \right) = E$

$$E = \frac{\hbar^2 \pi^2}{2\mu} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right)$$

$$E = \frac{h^2 \pi^2}{2m} \left[\left(\frac{n_x}{L_x} \right)^2 + \left(\frac{n_y}{L_y} \right)^2 + \left(\frac{n_z}{L_z} \right)^2 \right]$$

(Phase space): (q, p) : Dual to TM i.e. $(TM)^*$ or T^*M



$$\delta q \delta p = h_0 \leftrightarrow \text{constant}$$

Hazenberg: $\delta q \delta p \geq h$

Meaning choosing $h_0 < h \Rightarrow \delta q \delta p \geq h$ is meaningless as allowed by QM.

* $f: (q_1, \dots, q_f) \oplus (p_1, \dots, p_f)$
 So a total of $2f$ parameters
 $\dim(T^*M) = 2f$

Cords: $(q_1, \dots, q_f, p_1, \dots, p_f) \leftarrow p_t \text{ in } (T^*M)$

f : D.O.F.

ex: For N point particles: each pt. has 3 P.O.F. and hence
 $f = 3N$

* Subdivide via cells $\delta q_1, \dots, \delta q_f, \delta p_1, \dots, \delta p_f$

$$(\text{Volume of a cell}) = h_0^f$$

(Classical system) $\rightarrow$ (Quantum Mech)

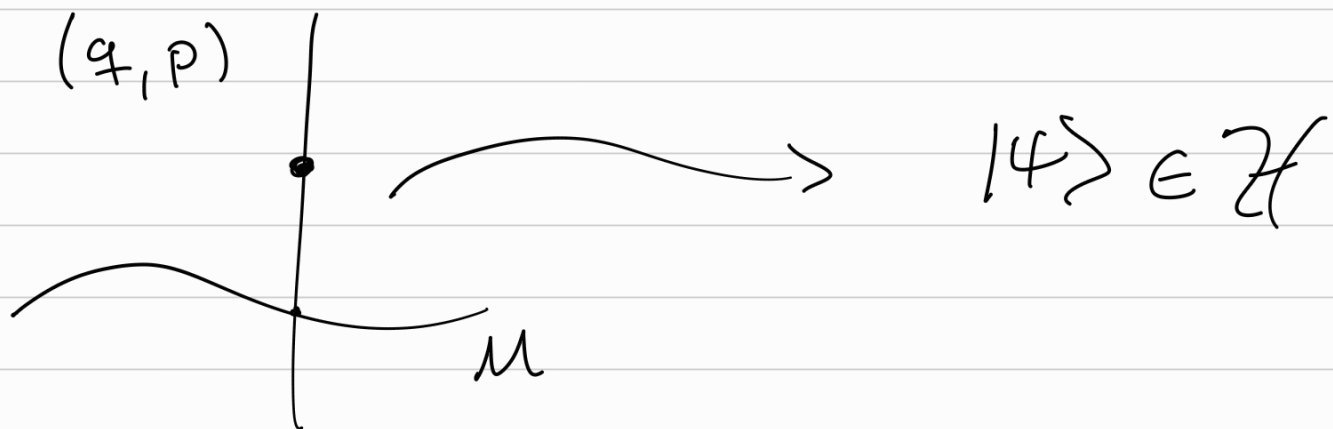
* Cell in T^*M

* Quantum State

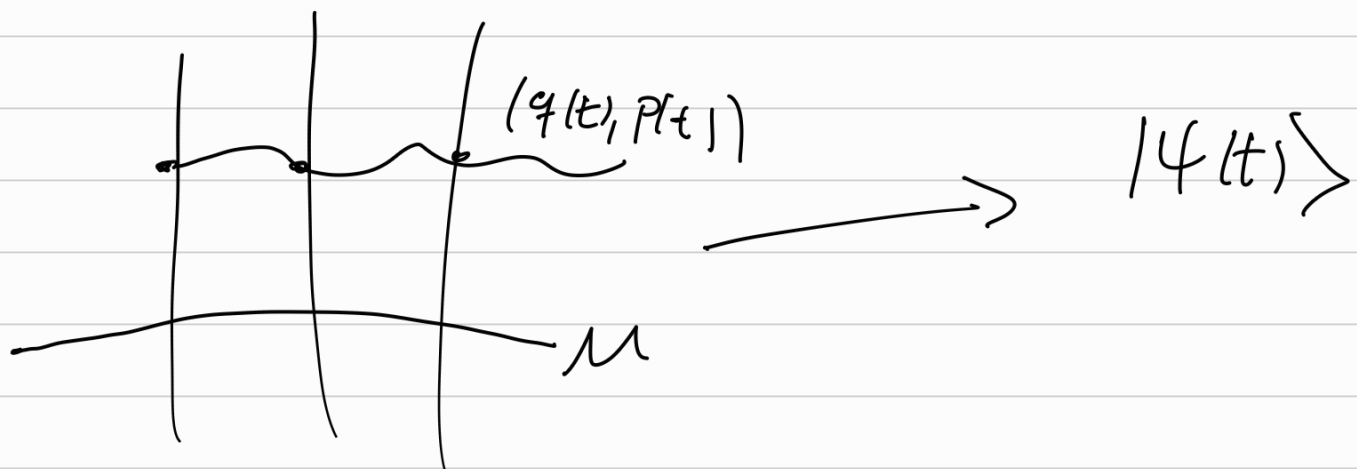
Statistical Ensemble:

QM: Knowledge of $|\psi\rangle$ allows us to det. $|\psi(t)\rangle$

CM: Knowledge of (q, p) we can determine $(q(t), p(t))$



$(q(t), p(t))$ $T^*M \cong F \times M$



* $\exists$ an iso? $\rho: T^*M \rightarrow TM' \cong M' \times \mathbb{C}$

$$\rho(q, p) = |\psi_{q,p}\rangle$$

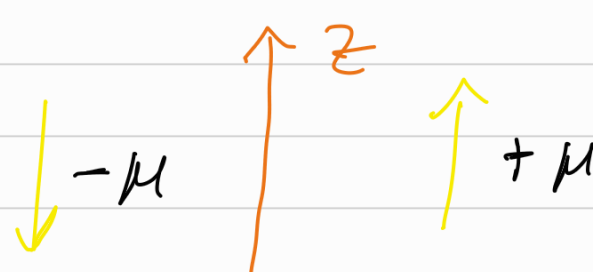
Q: How would the energies "translate" and

How would we determine these states.

ψ : Section of $TM \cong M \times \mathbb{P}$

Statistical Ensemble:

ex: 3 fixed particles, each w/ spin $\frac{1}{2}$ and so can have velocity ± 1 or -1 along $\uparrow z$

Magnetic Moment: 

$\leadsto$ External mag. field $\vec{H} = H \hat{z}$

(magnetic quantum number): $M_z = \pm \frac{1}{2}$

• State of the entire system is specified by (m_1, m_2, m_3)

$\uparrow +\mu \Rightarrow U_{\uparrow} = -\mu H$

$\downarrow -\mu \Rightarrow U_{\downarrow} = \mu H$

State label	(m_1, m_2, m_3)	$\sum M$	$\sum U_i$
1	+++	3μ	$-3\mu H$
2	++-	μ	$-\mu H$
3	+ - +	μ	$-\mu H$
4	- ++	μ	$-\mu H$
5	+ --	$-\mu$	μH
6	- + -	$-\mu$	μH
7	-- +	$-\mu$	μH
8	---	-3μ	$3\mu H$

- If we know the total energy:

$$\sum U_i = -\mu H$$

then we know the system is either in $(++-), (+-+), (-++)$

Postulates:

(i) Assume the system is isolated.

$\Rightarrow$ Total energy conserved

(ii) System is in equilibrium:

characterised by the fact that the prob. of finding the system in any one state is independent of time.

$$\frac{d}{dt} P(x) = 0 \quad \forall x$$

$\Rightarrow$ All macroscopic parameters are also time independent:

ex: $\frac{dV}{dt} = 0, \quad \frac{dN}{dt} = 0$

(iii) Assume each state is equally likely in terms of $\sum P_i = 1$ (until further info is given)
 $\rightarrow$ "Liouville's Theorem"

Postulate I: An isolated system in equilibrium is equally likely to be in any of its accessible states

$\rightarrow$ In CM: state $\leftrightarrow$ cell in T^*M

ex: (i) $(++-), (+-+), (-++)$

all are equally likely $\rightarrow$ each has $1/3$

- Does not apply to individual spins

(ii) N-atoms: $N \sim 6 \times 10^{23}$

• Same as above but $w/3 \rightarrow N$

(iii) 1D - harmonic Oscillator:

$$T^*M : (x, p)$$

$$E = \dot{p}^2/2m + \frac{1}{2}Kx^2$$

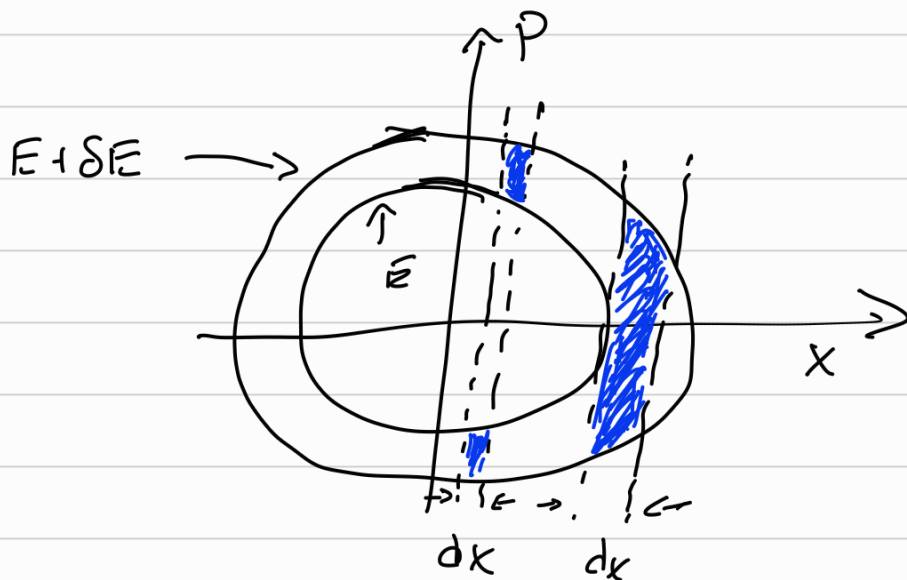
for constant E

$$\tilde{p} = p/\sqrt{2m}, \quad \tilde{x} = \sqrt{\frac{K}{2}} x$$

$$E = \tilde{p}^2 + \tilde{x}^2$$

→ Suppose the energy lands in the range between

$$E \text{ and } (E + \delta E)$$



$$x = A \cos(\omega t + \varphi)$$

$$|\omega = \sqrt{K/m}|$$

$$p = m \dot{x} = -mA\omega \sin(\omega t + \varphi)$$

$$E = \frac{m\omega^2}{2} A^2 \sin^2(\omega t + \varphi) + \frac{K}{2} A^2 \cos^2(\omega t + \varphi) = \frac{1}{2} m \omega^2 A^2$$

$$A = \sqrt{\frac{2E}{m\omega^2}}$$

The approach to Equilibrium:

* Not equal states → changes w/ time

→ These are non-equilibrium systems